EDNS 491 & 492: Capstone Design I & II

Student Guide

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How to use this guide: This document is your sprint-by-sprint companion for Capstone Design. It is *not* a replacement for the Course Syllabus, the Course Text, or Canvas; it is a navigation aid that tells you where to focus your energy each sprint. For each sprint, start by reading that sprint's section and pay special attention to any box labeled "Common Pitfall" or "Pro Tip" (those address the most frequent issues). Read the relevant section at the *start* of each sprint, not the end.

Three documents work together: (1) this Student Guide (sprint-level navigation), (2) the Course Text (deep reference, 22 chapters across 5 parts), and (3) Canvas (assignments, rubrics, deadlines). The Course Syllabus is the authoritative source for grading and policies.

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1 The Capstone Sprint: A Typical Week

This course meets formally on Tuesdays and Thursdays only (lectures and recitations). The Monday, Wednesday, and Friday activities shown below are *self-directed* team work sessions that you are responsible for organizing outside of class time. Capstone requires significant independent effort beyond scheduled meetings. Teams that rely solely on Tuesday/Thursday class time will fall behind.

Attendance at every lecture and recitation is mandatory. Key deliverable instructions, client feedback, design review logistics, and PA assessments occur during these sessions. Material covered in lecture and recitation cannot always be recovered from slides alone. Missing a session means missing information your team needs to stay on track.

Your team's weekly rhythm should follow this general pattern (adapt to your specific schedule):

Day	Activities
Monday	*Self-directed*: Team stand-up (15 min). Update task board. Identify blockers.
Tuesday	**Scheduled**: Lecture / Recitation / Lab session. Design, analysis, or fabrication work.
Wednesday	*Self-directed*: Work session. PA check-in (show progress, surface issues).
Thursday	**Scheduled**: Lecture / Recitation / Lab session. Send client status email (biweekly).
Friday	*Self-directed*: Work session. Documentation. End of Week 2: Sprint Review + Retrospective.

The three non-negotiable weekly touchpoints: (1) the stand-up (team coordination), (2) the PA check-in (accountability), and (3) the client communication (professionalism). Everything else flexes around your project needs. Plan for 6–10 hours per week of team work beyond scheduled class time.

2 Course Philosophy and the Three-Document Ecosystem

Capstone Design is not another course; it is a professional practice experience compressed into two semesters. You will define problems, manage a team, communicate with a real client, design and build (or model) a system, verify it against requirements, and present your results publicly at the Design Showcase. The skills you develop here (engineering judgment, project management, client communication, teamwork) are the skills employers value most.

What level of independence is expected?

Capstone demands self-motivation. Unlike lecture-based courses, your progress depends on work you organize and complete *outside* of scheduled Tuesday/Thursday class sessions. Your PA will not chase you for deliverables. Your client will not remind you of deadlines. You and your teammates are responsible for holding each other accountable, scheduling your own work sessions, and maintaining momentum between class meetings. Treat this course the way you would treat your first engineering job: your supervisor sets the objectives, but you manage your own time to meet them.

2.1 How SDI and SDII Fit Together

SDI (EDNS 491) is the **definition semester**: you go from project assignment through requirements, concept selection, and Preliminary Design Review. Sprints 0–6 cover the left side of the V-Model.

SDII (EDNS 492) is the **execution semester**: you re-engage with the client, complete the Critical Design Review, build/model your system, test it, and present at the Final Design Review and Design Showcase. Sprints 7–14 cover the bottom and right side of the V-Model.

2.2 Your Three Documents

- **This Student Guide**: tells you what to focus on each sprint.
- **The Course Text**: provides the engineering methodology, process guidance, and technical reference. Organized by topic in 22 chapters across 5 parts.
- **Canvas**: assignment rubrics, submission portals, professional development modules, scheduling, and the Labriola InnoHub enrollment site.

3 The Most Common Mistake at Each Sprint

Sprint	Most Common Mistake	Prevention
0	Starting to design before understanding the problem	Spend Sprint 0 listening, not solving. Ask questions, take notes.
1	SOW with no "Out of Scope" section	Explicitly list what you will NOT do. Get client agreement.
2	Generating only one concept	Use brainstorming, SCAMPER, morphological chart. Generate $\geq$15 ideas.
3	Decision matrix rigged to support a predetermined choice	Add sensitivity analysis. Vary weights by $\pm$20%.
4	PoC that does not retire the key technical risk	Identify the risk first, then design the PoC to test it specifically.
5	Skipping the Mock PDR	Schedule the mock 3 days before. Fresh eyes catch problems you miss.
6	Not ordering long-lead components	Order PCBs and specialty parts NOW. They arrive in time for SDII.

Sprint	Most Common Mistake	Prevention
7	Skipping the Hazard Assessment	Required before fabrication. PA must approve. No shortcuts.
8	BOM missing fasteners and consumables	Walk through the CAD assembly. Every visible component needs a BOM line.
9–10	Big-bang integration	Test each subsystem individually before connecting one at a time.
11	Leaving all testing to Sprint 12	Start V&V testing NOW. Every week of margin is one redesign cycle.
12–13	Documentation debt	Write report sections during the sprint the work is done, not at the end.
14	Incomplete VCRM at FDR	Every requirement must be dispositioned: Verified, Failed, Waived, or Deferred.

4 SDI Sprint 0: Project Expo, Team Formation, Client Kickoff

Weeks 1–2 | Course Text: Chapters 1, 2

▶ Top Priorities This Sprint

1. Attend the Project Expo and complete the Project Selection Survey.
2. Meet your team, PA, and client.
3. Prepare for and conduct the Client Kickoff Meeting.
4. Set up the team repository (Teams group, file structure).
5. Begin the Individual Design Log.

What should I do before the kickoff meeting?

Research the client's organization and industry. Prepare a slide deck for project discovery (Sprint 0 deliverable). Write 10 open-ended questions. Assign roles: facilitator, scribe, timekeeper. See Course Text Ch. 2 for the full kickoff workflow.

Do not start designing during Sprint 0. Your job this sprint is to *understand the problem*, not to solve it. Listen more than you talk at the kickoff meeting. The first idea is rarely the best idea.

Send a follow-up email to the client within 24 hours of the kickoff meeting summarizing key decisions, open questions, and next steps. Copy your PA. This establishes your professionalism from Day 1.

4.1 Deliverables

- What to Expect from SD Quiz (Canvas)
- Project Selection Survey (by 11:30 AM on the designated date)
- Prof. Dev. – IP Module, Licensure Module
- Email to client to introduce team with team slide

5 SDI Sprint 1: The Project Plan, Scrum Training, Client Meeting Prep

Weeks 3–4 | Course Text: Chapters 7, 11, 13

▶ Top Priorities This Sprint

1. Develop the Statement of Work (SOW) with scope, deliverables, schedule, budget, and roles.
2. Establish the Product Backlog and first Sprint Backlog.
3. Set up task tracking (Trello, Planner, or GitHub Projects).
4. Complete Scrum training modules on Canvas.
5. Create the Team Charter.

How detailed should the SOW be?

Detailed enough that all parties agree on what "done" looks like for every deliverable. Include explicit "Out of Scope" items. Your PA evaluates the project against the SOW. See Course Text Ch. 7 for the complete SOW structure.

The most common Sprint 1 mistake is skipping the "Out of Scope" section. If you don't document what you will *not* do, the client may assume it's included.

5.1 Deliverables

- Prof. Dev. – Intro to Scrum, Creating Team Culture Reading
- Client Kickoff Meeting slides and meeting minutes
- Initial Product Backlog
- EHS Safety Training (Canvas module)

6 SDI Sprint 2: Concept Generation, Trade Study, Career Day

Weeks 5–6 | Course Text: Chapters 4, 20

▶ Top Priorities This Sprint

1. Generate $\geq$15 concepts using brainstorming, SCAMPER, and morphological charts.
2. Screen concepts with Pugh matrix; advance top 3–5 to decision matrix.
3. Prepare for Career Day (resume, elevator pitch, professional attire).
4. Sprint 1 Review / Retrospective / Sprint 2 Backlog.

Career Day is a prime networking opportunity. Bring 10–15 printed resumes and practice your capstone elevator pitch. See Course Text Ch. 20 for Career Day preparation.

Do not settle on a concept before generating alternatives. Three variations of the same idea are *not* three concepts. Generate genuinely different approaches before converging.

6.1 Deliverables

- Concept generation evidence (sketches, morphological chart)
- Prof. Dev. – Adv. Scrum Training
- Sprint 1 Review / Retrospective documentation
- Project Plan emailed to client

7 SDI Sprint 3: Concept Development, Client Approval of Approach

Weeks 7–8 | Course Text: Chapters 3, 4, 17

▶ Top Priorities This Sprint

1. Complete the weighted decision matrix with sensitivity analysis.
2. Present top 2–3 concepts to client; obtain approval of selected approach.
3. Draft the SDRD with initial requirements.
4. Identify applicable engineering standards.
5. Sprint 2 Review / Retrospective / Sprint 3 Backlog.

What if the client doesn't like any of our concepts?

Ask which aspects they do like and which concern them. Use their feedback to generate a hybrid concept or explore a new direction. Client approval is a scope commitment; invest the time to get it right.

7.1 Deliverables

- Decision matrix with sensitivity analysis
- Client approval of proposed solution (documented in meeting minutes)
- Engineering Standards identification
- Ethics Assignment / Peer Evaluation #1

8 SDI Sprint 4: Functional Prototypes / Site Plans, Ethics Assignment

Weeks 9–10 | Course Text: Chapters 5, 12, 22

▶ Top Priorities This Sprint

1. Build proof-of-concept prototype or develop analytical model.
2. Complete the SDRD (all requirements with TAID and MoSCoW).
3. Apply CLO 5 sustainability tool (LEED, ENVISION, DFE, or ESJ).
4. Complete the Ethics Assignment.
5. Sprint 3 Review / Retrospective / Sprint 4 Backlog.

Build Track: the PoC must demonstrate at least one physical function. "We drew it in CAD" is not a PoC. Paper Track: the PoC must demonstrate computational goal attainment (e.g., converging simulation, validated sub-model).

If using InnoHub shops, start machine training *now*. Training slots fill up quickly and you cannot use equipment without completing the required training on the Labriola Canvas site.

8.1 Deliverables

- Proof-of-Concept demonstration (photos/video)
- Ethics Assignment (1000–1500 word argumentative essay)
- Complete SDRD draft

9 SDI Sprint 5: PDR Preparation, Mock PDR, PDR Execution

Weeks 11–13 | Course Text: Chapters 6, 8, 18, 19

▶ Top Priorities This Sprint

1. Finalize PDR Report (PDRt) and send to client in advance.
2. Conduct Mock PDR (practice presentation with PA or another team).
3. Deliver PDR presentation (PDRw) to PA, client, and TAs.
4. Populate the VCRM (plan: all entries "Open" with TAID assigned).
5. Sprint 4 Review / Retrospective / Sprint 5 Backlog.

What does a strong PDR look like?

Strong PDRs have: testable requirements, $\geq$3 distinct concepts, a decision matrix with sensitivity analysis, proof-of-concept evidence (not just claims), a sustainability assessment, and a team that can answer questions about every section. See Course Text Ch. 8 for the full checklist and common failure patterns.

Do not skip the Mock PDR. Teams that rehearse consistently outperform teams that present cold. Schedule the mock at least 3 days before the formal review.

9.1 Deliverables

- PDR Report (PDRt) submitted to PA and client
- PDR Presentation (PDRw) delivered
- VCRM plan
- Individual Performance Eval #1

10 SDI Sprint 6: Design Refinement, Next Steps, SDII Handoff

Weeks 14–15 | Course Text: Chapters 8, 10

▶ Top Priorities This Sprint

1. Incorporate PDR feedback: categorize, assign owners, complete.
2. Baseline the SDRD (freeze; all future changes under change control).
3. Begin procurement of long-lead-time components for SDII.
4. Prepare SDII transition documentation.
5. Send Next Steps Email to client.

Order long-lead-time components *now*. Do not wait until SDII Sprint 8. Custom PCBs take 2–4 weeks. Specialty sensors take 2–4 weeks. Every week you wait in Sprint 6 is a week less margin in SDII.

10.1 Deliverables

- PDRt Update (incorporating feedback)
- Next Steps Email to client
- Peer Evaluation #2
- End-of-Semester Course Survey
- Sprint 5 Review / Retrospective / Sprint 6 documentation

11 SDII Sprint 7: Re-engagement, Goal Setting, Hazard Assessment

Week 1 | Course Text: Chapters 2, 11, 21

▶ Top Priorities This Sprint

1. Send Re-engagement Email to client (5 required elements: reintroduce team, summarize SDI status, propose SDII goals, request meeting, attach Group Reflection).
2. Complete the Hazard Assessment and Safety Plan.
3. Review and update the Product Backlog with client.
4. Set SDII semester goals with PA and client.

The Hazard Assessment and Safety Plan must be approved by your PA *before* fabrication begins. Do not start building without it. See Course Text Ch. 21 for the template and required content.

11.1 Deliverables

- Re-engagement Email and Group Reflection & Action Plan
- EHS Safety Training (SDII module)
- Hazard Assessment & Safety Plan

12 SDII Sprint 8: CDR Report, Eng Calc Package, CDR Execution

Weeks 2–5 | Course Text: Chapters 6, 9, 14, 15

▶ Top Priorities This Sprint

1. Finalize CDR Report (CDRt) and Engineering Calculation Package.
2. Send CDRt to client before CDR presentation.
3. Deliver CDR presentation (CDRw) to PA, client, and TAs.
4. Complete BOM with all components ordered or received.
5. Complete InnoHub Manufacturing Review Meeting (required before ordering build materials).

The Manufacturing Review Meeting is a **required checkoff on the purchasing form**. CP will not process material orders until it is complete. Schedule it early; slots fill up around CDR time.

Freeze the design at CDR. After this point, changes require formal change control. Teams that keep redesigning after CDR arrive at FDR with no working prototype.

12.1 Deliverables

- CDR Report (CDRt) with Engineering Calculation Package
- CDR Presentation (CDRw)
- Peer Evaluation #1 (SDII)
- Next Steps Email (post-CDR) to client

13 SDII Sprints 9–11: Build/Test/Iterate, Broader Impacts Essay

Weeks 6–11 | Course Text: Chapters 5, 6, 12

▶ Top Priorities This Sprint

1. Fabricate, assemble, and wire your system (Build Track) or develop full-fidelity models (Paper Track).
2. Test subsystems individually *before* integrating (test each subsystem in isolation, then integrate one at a time).
3. Begin system-level integration testing.
4. Write the Broader Impacts Essay (SDII, 1000–1500 words argumentative).
5. Sprint Reviews / Retrospectives every 2 weeks.

The Spring Break Shipping Trap: campus is closed for a full week during Spring Break. If a component fails during testing and you need a replacement, order it *before* break. Plan all critical procurement to arrive at least one week before break starts.

Start subsystem testing as early as Sprint 9. Do *not* wait until Sprint 12 for your first test. Every week of testing margin you have is a week of margin for redesign if something fails.

13.1 Deliverables

- Broader Impacts Essay
- Mid-Semester Course Survey
- Peer Evaluation #2 (SDII) / Individual Performance Eval #1
- Sprint Review / Retrospective documentation (each sprint)

14 SDII Sprints 12–13: FDR Preparation, System V&V, Final Testing

Weeks 12–14 | Course Text: Chapters 6, 10, 18

▶ Top Priorities This Sprint

1. Complete all verification testing and update the VCRM.
2. Disposition any failed requirements (redesign, waiver, risk acceptance).
3. Write the FDR Report (FDRt) including as-designed vs. as-built comparison.
4. Write the O&M manual.
5. Design and submit the Showcase poster for printing (∼1 week before Showcase).

A VCRM with mostly "Open" entries at FDR indicates insufficient testing. Every requirement must be dispositioned: Verified, Failed (with disposition), Waived, or Deferred. See Course Text Ch. 6 for failure disposition options.

Write the O&M manual for a non-engineer. Use photographs, numbered steps, and clear language. The person who operates your system after you graduate may not have your technical background.

14.1 Deliverables

- FDR Report (FDRt) submitted to PA and client
- Complete VCRM (all entries dispositioned)
- O&M manual
- Showcase poster (submitted for printing)
- Design-Build-Test-Redesign Slide

15 SDII Sprint 14: FDR, Showcase, Documentation, Final Checkout

Weeks 15–16 | Course Text: Chapters 10, 19

▶ Top Priorities This Sprint

1. Deliver FDR presentation (FDRw) to PA, client, and TAs.
2. Present at the Design Showcase (poster + live demonstration).
3. Complete all final deliverables.
4. Final Checkout with PA.

What is Final Checkout?

A closing meeting with your PA to: hand off all project materials, return borrowed equipment, submit the Digital Design Archive, review your Project Goal Attainment, and formally close the project. Schedule it before the last day of class.

15.1 Final Deliverables

- FDR Presentation (FDRw)
- Digital Design Archive (all project files organized)
- "What I Wish I Would Have Known" team video
- Project Synopsis
- Next Steps Email (post-FDR) to client
- Project Goal Attainment
- Peer Evaluation #3 / Individual Performance Eval #2
- Final Course Survey
- Final Checkout with PA

16 Design Review Preparation Checklist

Reference | Course Text: Chapters 8, 9, 10

Before every design review, complete the **Project Health Check** self-assessment (Course Text, Appendix B). Rate your team Green/Yellow/Red on each of the 12 failure patterns. Discuss any Yellow or Red ratings with your PA.

1 week before:

- Report draft complete and under team review.
- Slide deck outline complete with section owners.
- PoC / demo hardware tested and working.

3 days before:

- Report finalized and sent to client.
- Slides complete; first rehearsal conducted.
- Identified and researched likely Q&A questions.

Day of:

- Arrive 15 minutes early; test projector and demo equipment.
- Backup slides on USB drive.
- Assigned Q&A responders by topic.
- Scribe ready to record feedback.

17 What Your PA Expects

Reference | Course Text: Chapters 11, 13, 17

Your PA evaluates you based on both *what* you deliver and *how* you deliver it. Meeting the checklist is necessary but not sufficient.

What separates an A team from a B team?

A teams: communicate proactively (PA hears about problems before they become crises), incorporate feedback aggressively (PDR weaknesses are addressed by CDR), demonstrate engineering judgment (trade-offs are analyzed with data, not guesses), and produce professional-quality documentation (reports read like they were written by engineers, not by students rushing at midnight).
B teams: meet deadlines but do not exceed expectations, address feedback partially, make design decisions without data, and produce adequate but unpolished documentation.

The single most effective way to improve your PA's assessment: **close the feedback loop**. When your PA gives feedback at a sprint review, write it down, address it, and report back at the next meeting: "At the last review you asked about our thermal analysis. Here are the results." PAs notice when their feedback is taken seriously.

18 The Feedback Loop Survival Guide

Reference | Course Text: Chapter 13

Feedback Loop peer evaluations directly affect your individual grade. SDI has 2 evaluations; SDII has 3.

There are two types of individual assessment: **Peer Evaluations** (completed by your teammates about each other) and **Individual Performance Evaluations** (completed by your PA about you). Both inform your individual grade. Check Canvas for the specific schedule of each.

A pattern of low Feedback Loop scores (below 3.0 out of 5.0) combined with PA observation of non-participation can result in individual grade reduction, PIP initiation, or course failure regardless of team scores.

How to receive good evaluations: attend every meeting on time, complete your tasks by the agreed deadline, communicate proactively when you are behind, help teammates when you finish early, participate actively in discussions (not just silently), and be responsive to messages within 24 hours.

How to give useful evaluations: be specific. "Jordan was great" does not help the PA calibrate. "Jordan completed the firmware ahead of schedule, debugged the I2C communication issue that was blocking the sensor team, and volunteered to write the CDR power analysis section when the original author was overwhelmed" gives the PA evidence to work with.

Self-evaluation: be honest. PAs cross-check your self-rating against peer ratings and their own observations. Over-rating yourself while peers rate you lower signals a lack of self-awareness. Acknowledging a specific weakness ("I was late on two deliverables in Sprint 10 due to competing course deadlines; I should have communicated the conflict earlier") signals maturity.

19 CFO and Purchasing Survival Guide

Reference | Course Text: Chapters 14, 15

The CFO role is one of the most time-consuming in SDII. Here is the essential workflow:

1. **Get PA approval** before *every* purchase.

2. Check the **approved vendor list** on Canvas.
3. Complete the **Manufacturing Review Meeting** before ordering build materials (required checkoff).
4. Submit the correct form to CP: Purchase Form, Shopping List, or Reimbursement Form.
5. Email CP (capstonepurchasing@mines.edu) with: team name/number, date, vendor, form used.
6. Update the **Budget Tracker** monthly and upload to Canvas by deadline.
7. Pick up packages promptly from Labriola 204.
8. Use the Mines **tax exemption certificate** for all purchases (taxes are not reimbursed).

Purchases made without PA approval may result in **personal financial liability**. Always get approval first.

20 Travel Planning Quick Reference

Reference | Course Text: Chapter 16

- PA must approve Go/No-Go with CLT before any spending.
- Prepare a draft budget before meeting with CP.
- Carpooling required; multiple drivers same trip not reimbursed.
- Mileage: Mines rate, from 1500 Illinois, Golden CO 80401. Save Google Maps PDF.
- Meal receipts: itemized, no alcohol (entire receipt rejected).
- Submit all receipts within one month of trip.

21 Resume and Career Day Preparation

Reference | Course Text: Chapter 20

- Update your resume with capstone contributions (quantify: budget managed, components designed, tests conducted).
- Practice your 60-second elevator pitch.
- Prepare 5–7 STAR stories from capstone for behavioral interviews.
- Dress business casual; bring 10–15 printed resumes.
- Connect with PA, client, and TA on LinkedIn after the project.

22 Module Checklist: Sprint-to-Chapter Mapping

Sprint	Weeks	Primary Activities	Course Text Chapters
SDI (EDNS 491)			
0	1–2	Project Expo, team formation, client kickoff	Ch. 1, 2

Sprint	Weeks	Primary Activities	Course Text Chapters
1	3–4	Project Plan, Scrum training, product backlog	Ch. 7, 11, 13
2	5–6	Concept generation, trade study, Career Day	Ch. 4, 20
3	7–8	Concept development, client approval	Ch. 3, 4, 17
4	9–10	Prototypes / site plans, ethics assignment	Ch. 5, 12, 22
5	11–13	PDR preparation, mock PDR, PDR execution	Ch. 6, 8, 18, 19
6	14–15	Design refinement, next steps, SDII handoff	Ch. 8, 10
SDII (EDNS 492)			
7	1	Re-engagement, goal setting, safety plan	Ch. 2, 11, 21
8	2–5	CDR report, eng. calc package, CDR execution	Ch. 6, 9, 14, 15
9–10	6–8	Build/test/iterate, Broader Impacts essay	Ch. 5, 6, 12
11	9–11	System integration, V&V execution	Ch. 6
12–13	12–14	FDR preparation, final testing, documentation	Ch. 10, 18
14	15–16	FDR, Showcase, archive, final checkout	Ch. 10, 19

23 Quick Reference: Capstone Abbreviations

Abbr.	Meaning	Abbr.	Meaning
BOM	Bill of Materials	O&M	Operations & Maintenance
CDR	Critical Design Review	PA	Project Advisor
CF	Course Faculty	PDR	Preliminary Design Review
CLO	Course Learning Outcome	PIP	Performance Improvement Plan
CLT	Capstone Leadership Team	PoC	Proof of Concept
CP	Capstone Purchasing	Prof. Dev.	Professional Development
EHS	Environmental Health & Safety	SDI	Senior Design I (EDNS 491)
FDR	Final Design Review	SDII	Senior Design II (EDNS 492)
FMEA	Failure Mode & Effects Analysis	SDRD	System Design Req. Document
ICD	Interface Control Document	SOW	Statement of Work
InnoHub	Labriola Innovation Hub	TAID	Test, Analysis, Inspection, Demo
MoSCoW	Must/Should/Could/Won't	VCRM	Verification Cross-Ref. Matrix
MVP	Minimum Viable Product		

24 Emergency Contacts and Key Resources

Resource	Contact
Emergency	911
Mines Police (non-emergency)	303-273-3333
Capstone Program Director	Dr. Kristy Csavina, kcsavina@mines.edu
Capstone Purchasing	capstonepurchasing@mines.edu, Labriola 204
InnoHub Operations	Julia Roos, jroos@mines.edu
InnoHub General	innohub@mines.edu
EHS	303-273-3316, https://www.mines.edu/ehs/
Writing Center	https://writing.mines.edu/, appointments at mines.mywconline.com
Labriola Info	http://labriola.mines.edu/

This guide is a companion to the EDNS 491/492 Course Text. For detailed methodology, technical reference, and worked examples, consult the Course Text chapters referenced in each section above.